

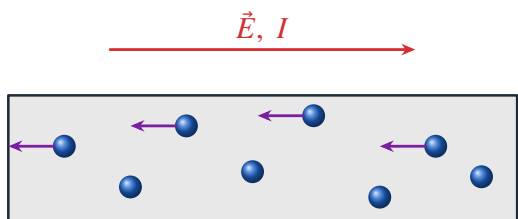
CURRENT ELECTRICITY

Current • Drift • Resistance

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1 Current and Drift Velocity

$$I = \frac{dq}{dt} = neAv_d$$

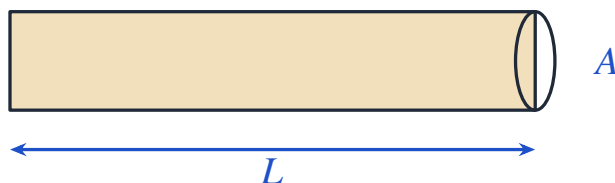


electrons drift against $\vec{E}$

$v_d \sim 10^{-4}$ m/s, yet the signal travels at nearly c .

2 Ohm's Law and Resistivity

$$V = IR, \quad R = \frac{\rho L}{A}$$



ρ depends on the material and temperature, not on the shape.

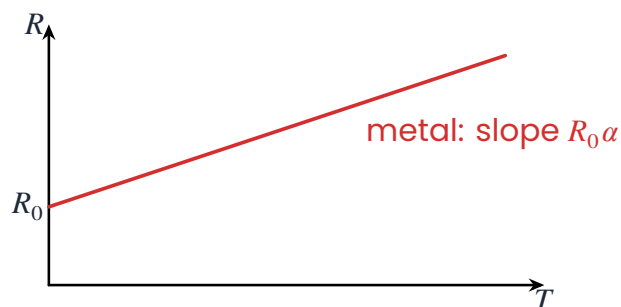
3 Stretching a Wire

$$L \rightarrow nL \Rightarrow R \rightarrow n^2R$$

Volume stays the same, so $A \rightarrow A/n$.
With radius $r \rightarrow r/n$: $R \rightarrow n^4R$.

4 Temperature

$$R_T = R_0 (1 + \alpha \Delta T)$$



Metals: $\alpha > 0$. Semiconductors: R falls as T rises.

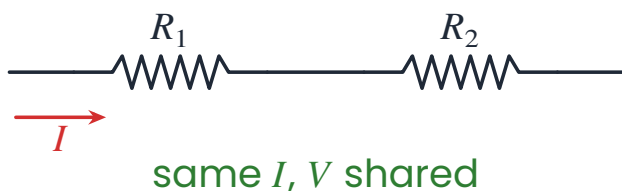
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Combinations • Real Cell

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5 Series

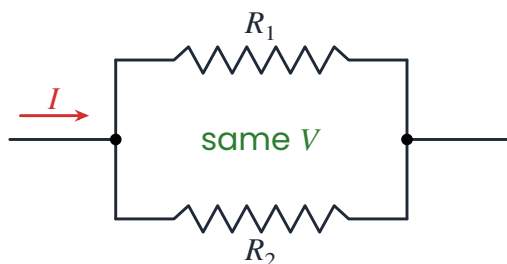
$$R_s = R_1 + R_2 + \dots$$



Same current; voltage divides in the ratio of the resistances.

6 Parallel

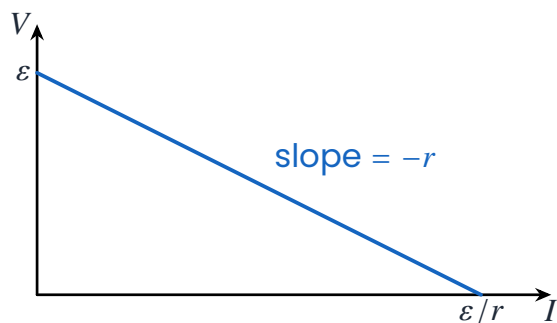
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$



Same voltage; current divides in the inverse ratio.

7 EMF and Internal Resistance

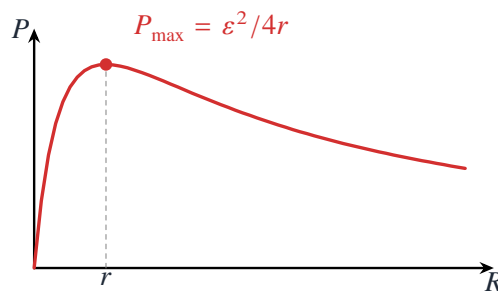
$$V = \varepsilon - Ir, \quad I = \frac{\varepsilon}{R + r}$$



While charging: $V = \varepsilon + Ir$.

8 Maximum Power

$$P = \frac{\varepsilon^2 R}{(R + r)^2}$$



Largest when $R = r$ (matched load).

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Grouping of Cells • Kirchhoff

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9 Cells in Series

$$I = \frac{n\varepsilon}{R + nr}$$

Best when $R \gg nr$.

10 Mixed Grouping

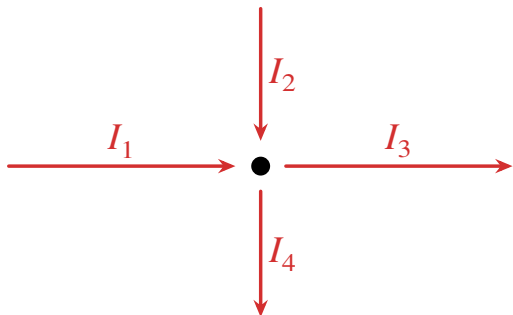
$$I = \frac{mn\varepsilon}{mR + nr}$$

largest when $R = \frac{nr}{m}$

n cells in each row, m rows in parallel.

11 Junction Law (KCL)

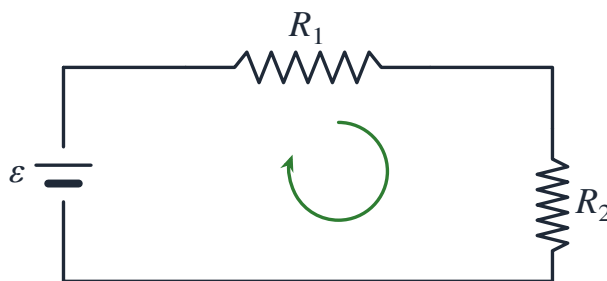
$$\Sigma I_{\text{in}} = \Sigma I_{\text{out}}$$



Here $I_1 + I_2 = I_3 + I_4$: charge is conserved.

12 Loop Law (KVL)

$$\Sigma \Delta V = 0 \text{ round a loop}$$



Across R along I : $-IR$. Across a cell – to $+$: $+\varepsilon$.

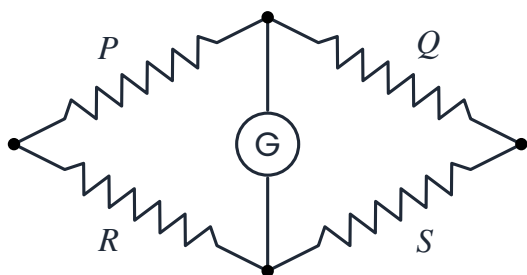
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Bridges • Power

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13 Wheatstone Bridge

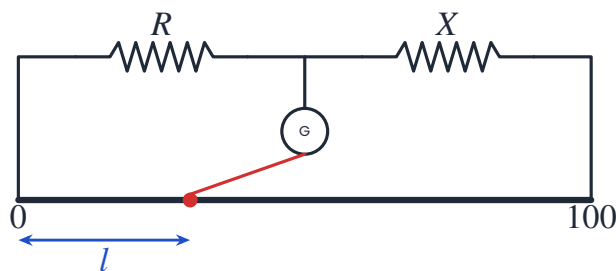
$$\frac{P}{Q} = \frac{R}{S} \Rightarrow I_G = 0$$



Balanced: remove the galvanometer arm, nothing changes.

14 Meter Bridge

$$\frac{R}{X} = \frac{l}{100 - l} \Rightarrow X = \frac{R(100 - l)}{l}$$



Swap R and X to cancel the end errors.

15 Electric Power

$$P = VI = I^2R = \frac{V^2}{R}$$

Rated (P_0, V_0) : $R = V_0^2/P_0$. Energy: 1 kWh
 $= 3.6 \times 10^6$ J.

16 Quick Numerical: Bulbs

100 W and 25 W bulbs (220 V) in series across 220 V. Which glows brighter?

$$R_{100} = 484 \, \Omega, \quad R_{25} = 1936 \, \Omega$$

$$P_{100} = 4 \, \text{W}, \quad P_{25} = 16 \, \text{W} : \text{the 25 W bulb}$$

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Quick Revision • Traps

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17 Everything at a Glance

Idea	Formula	Note
Current	$I = neAv_d$	$v_d = eE\tau/m$
Resistance	$R = \rho L/A$	stretch n times: $n^2 R$
Real cell	$V = \varepsilon - Ir$	$P_{\max}$ at $R = r$
Series / parallel	$R_1 + R_2 ; R_1 R_2 / (R_1 + R_2)$	same I ; same V
Bridge	$P/Q = R/S$	$I_G = 0$
Power	$I^2 R = V^2 / R$	series: lower watt glows more

18 Common Traps

- v_d is tiny; the current starts at once everywhere.
- Terminal voltage $\neq$ emf when current flows.
- In series the lower-wattage bulb glows brighter.
- A balanced bridge's middle arm carries no current.

19 Colour Code

Band	Meaning
1st, 2nd	digits
3rd	multiplier 10^n
Gold / Silver	$\pm 5\%$ / $\pm 10\%$